Advanced Programming Practice · 21CSC203P

Unit 1 · Session 12 · SLO 2 — Solved Worksheet

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Unit – 1 : Session – 12 : SLO – 2 | Serialization & Distributed Processing

Activity (Solutions)

• 1) class Student implements Serializable { String name; int marks; Student(String n, int m){ name=n; marks=m; } } Serialize: ObjectOutputStream o = new ObjectOutputStream(new FileOutputStream("s.dat")); o.writeObject(new Student("Sam",90)); o.close(); Deserialize: ObjectInputStream in = new ObjectInputStream(new FileInputStream("s.dat")); Student s = (Student) in.readObject();

• 2) serialVersionUID: a version number for the Serializable class; during deserialization the JVM checks that the stored serialVersionUID matches the current class's — mismatch throws InvalidClassException. Declaring private static final long serialVersionUID = 1L; keeps control over compatibility when the class evolves.

• 3) Serialized objects in file I/O: write a list of Students to a file with ObjectOutputStream (as above) so objects persist between runs or travel across a network (distributed processing); on restart, ObjectInputStream rebuilds the exact objects — enabling save/load and sending objects between JVMs.